# CMAN — Roofline Construction and Kernel Classification

ECE 410/510: Hardware for AI and ML, Spring 2026

## Hardware Specifications

| Parameter | Value |
|---|---|
| Peak Compute (P_peak) | 10 TFLOPS = 10,000 GFLOP/s |
| Peak DRAM Bandwidth (B_peak) | 320 GB/s |
| Ridge Point (I*) | P_peak / B_peak = 10,000 / 320 = 31.25 FLOP/byte |

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## Task 1: Roofline Construction (Log-Log Axes)

### Bandwidth Ceiling (diagonal)

The bandwidth-limited performance is defined as:

```
Performance = I x B_peak
```

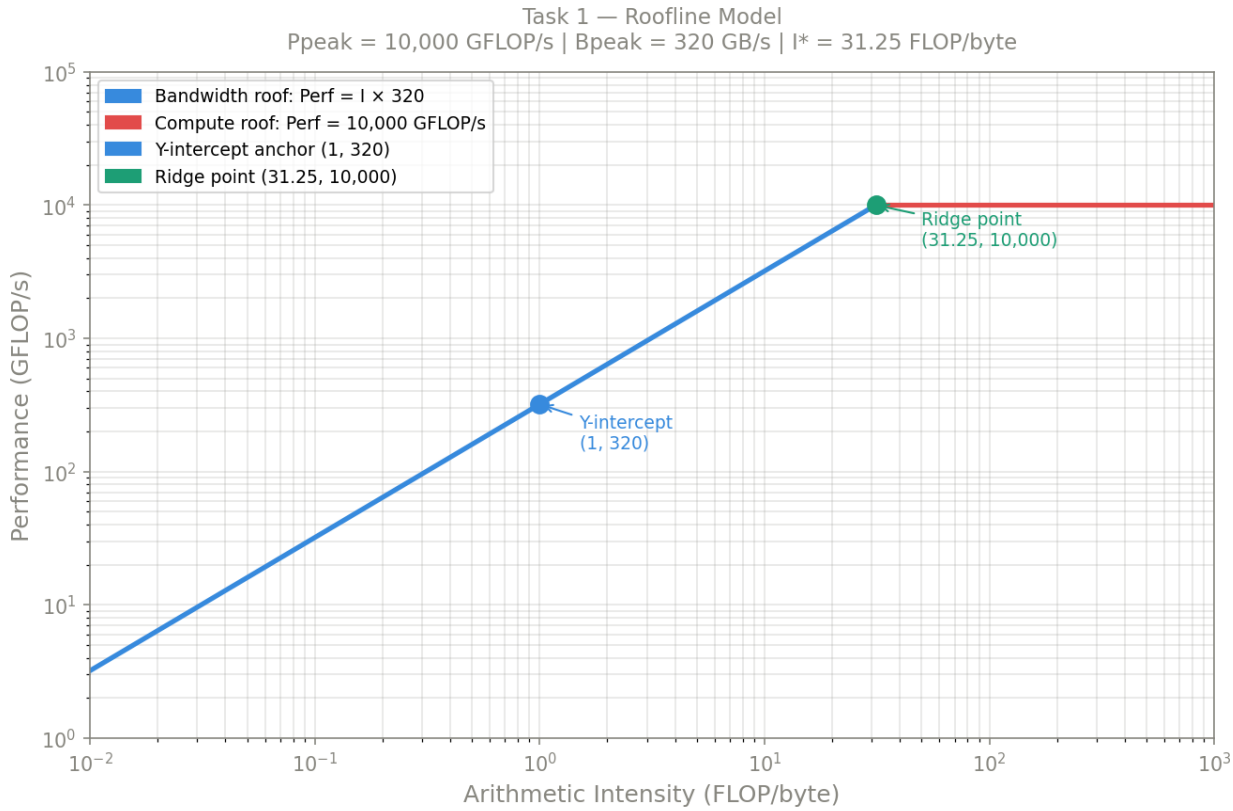
On log-log axes this becomes a straight line with **slope = 1**, anchored by two points:

| Point | Arithmetic Intensity | Performance | Justification |
|---|---|---|---|
| Y-intercept anchor | 1 FLOP/byte | 320 GFLOP/s | Perf = 1 x B_peak = 1 x 320 |
| Ridge point | 31.25 FLOP/byte | 10,000 GFLOP/s | I* = P_peak / B_peak = 10,000 / 320 |

### Compute Ceiling (flat horizontal)

At any I > 31.25 FLOP/byte, performance is capped at P_peak = 10,000 GFLOP/s.

### Roofline Plot


Task 1 — Roofline Model
Ppeak = 10,000 GFLOP/s | Bpeak = 320 GB/s | I* = 31.25 FLOP/byte
Bandwidth roof: Perf = I × 320
Compute roof: Perf = 10,000 GFLOP/s
Y-intercept anchor (1, 320)
Ridge point (31.25, 10,000)
Ridge point
(31.25, 10,000)
Y-intercept
(1, 320)
Performance (GFLOP/s)
Arithmetic Intensity (FLOP/byte)
$10^0$
$10^1$
$10^2$
$10^3$
$10^4$
$10^5$
$10^{-2}$
$10^{-1}$
$10^0$
$10^1$
$10^2$
$10^3$

## Task 2: Kernel A — Dense GEMM (1024x1024, FP32)

### FLOPs

Using formula $2N^3$ for square matrix multiply:

```
FLOPs = 2 x N^3 = 2 x 1024^3 = 2 x 1,073,741,824 = 2,147,483,648 FLOPs = 2.147 GFLOPs
```

### Bytes Transferred

Three matrices A, B, C each of size NxN, FP32 = 4 bytes each, no cache reuse:

```
Bytes = 3 x N^2 x 4 = 3 x 1024^2 x 4 = 3 x 1,048,576 x 4 = 12,582,912 bytes = 12.58 MB
```

### Arithmetic Intensity

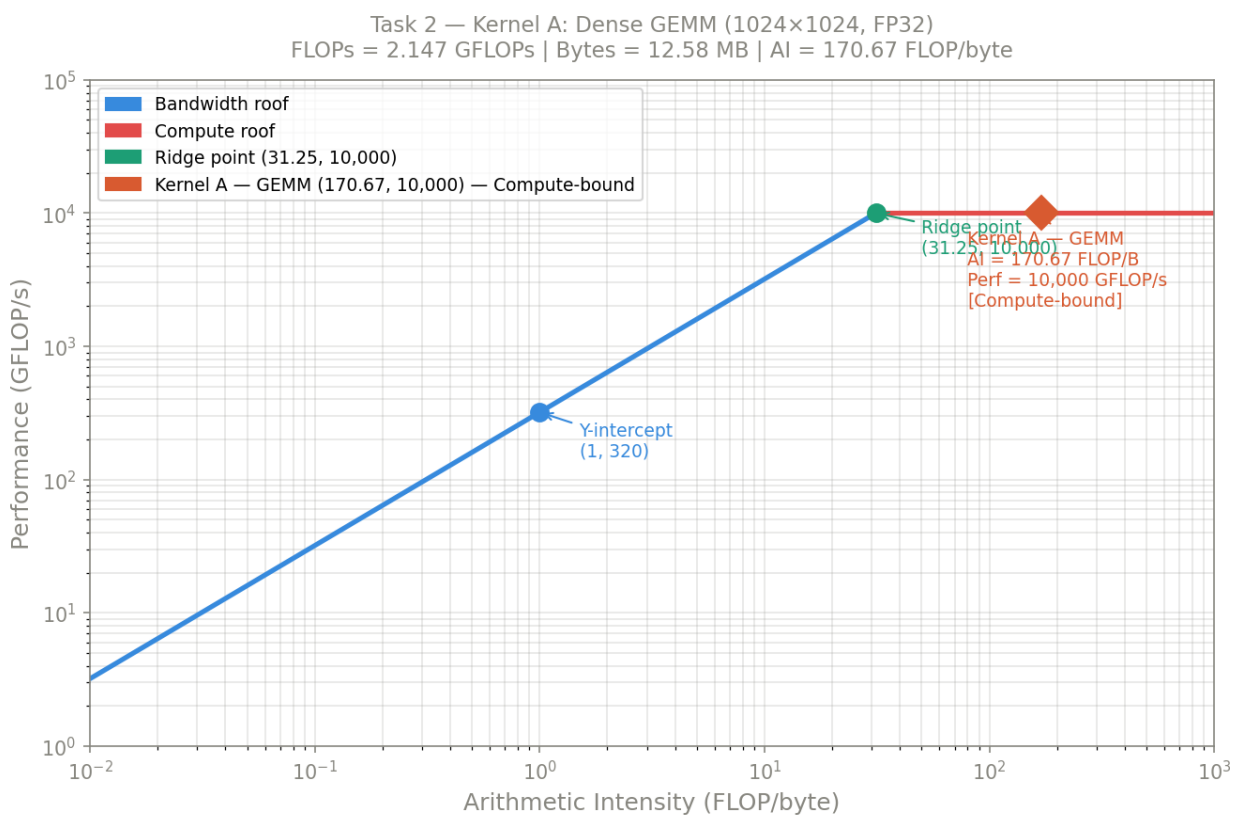
```
I = FLOPs / Bytes = 2,147,483,648 / 12,582,912 = 170.67 FLOP/byte
```

### Roofline Position

I = 170.67 FLOP/byte > I* = 31.25 FLOP/byte → **Compute-bound**

Attainable performance = min(P_peak, I x B_peak) = min(10,000, 170.67 x 320) = **10,000 GFLOP/s**

### Roofline Plot — Kernel A



## Task 3: Kernel B — Vector Addition (N = 4,194,304, FP32)

### FLOPs

One addition per element: C[i] = A[i] + B[i]

```
FLOPs = N = 4,194,304 FLOPs = 4.19 MFLOPs
```

### Bytes Transferred

Three arrays — read A, read B, write C — each length N, FP32 = 4 bytes:

```
Bytes = 3 x N x 4 = 3 x 4,194,304 x 4 = 50,331,648 bytes = 50.33 MB
```

### Arithmetic Intensity

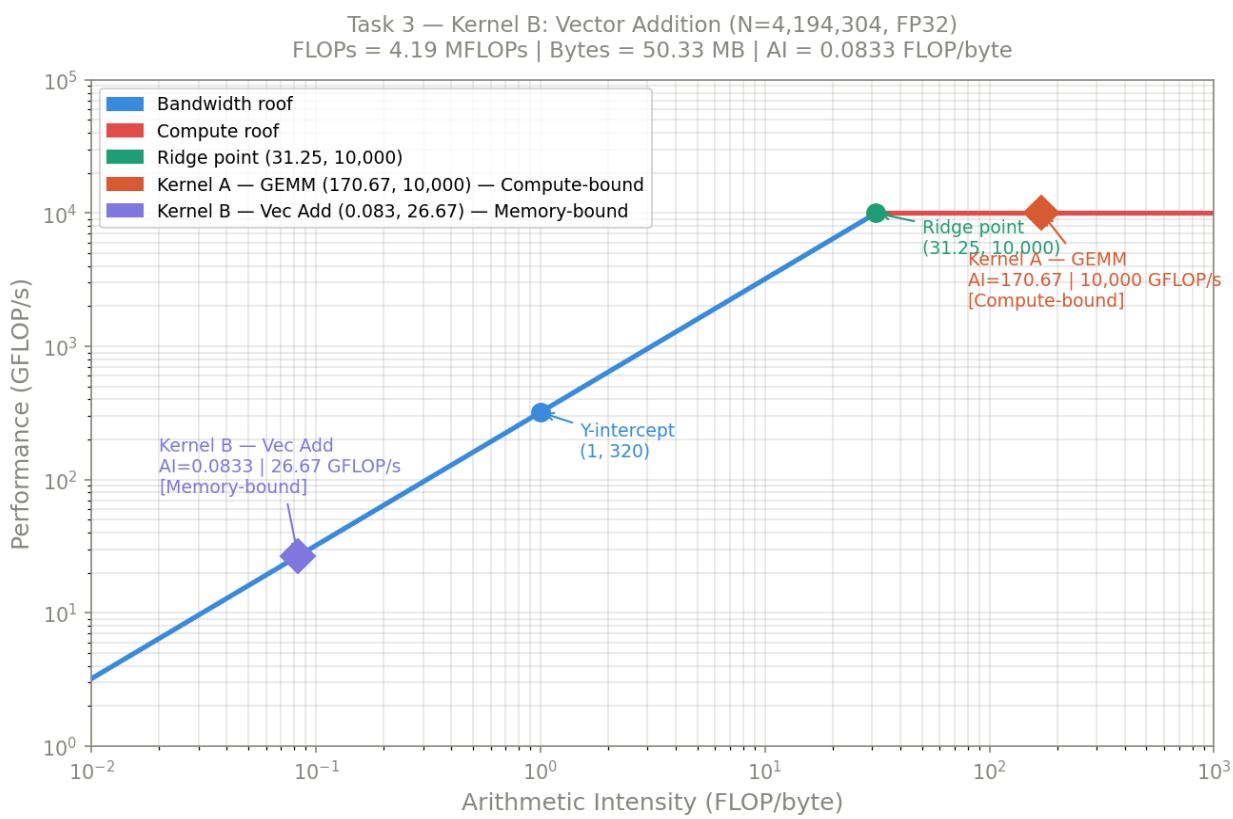
```
I = FLOPs / Bytes = 4,194,304 / 50,331,648 = 0.0833 FLOP/byte
```

### Roofline Position

I = 0.0833 FLOP/byte < I* = 31.25 FLOP/byte → **Memory-bound**

Attainable performance = min(P_peak, I x B_peak) = min(10,000, 0.0833 x 320) = **26.67 GFLOP/s**

### Roofline Plot — Kernel A + Kernel B



## Task 4: Kernel Analysis and Architectural Recommendations

### Kernel A — Dense GEMM

**(a) Bound classification:**

AI = 170.67 FLOP/byte > I* = 31.25 FLOP/byte → **Compute-bound**

**(b) Attainable performance ceiling:**

```
Perf = min(P_peak, I x B_peak) = min(10,000, 170.67 x 320) = 10,000 GFLOP/s
```

**(c) Architectural recommendation:**

Since GEMM is compute-bound, the bottleneck is raw compute throughput — not memory bandwidth. The most impactful architectural change is **increasing the number of MAC units or adding wider tensor core arrays** to raise P_peak. Increasing memory bandwidth would have no effect since the kernel is already compute-limited.

### Kernel B — Vector Addition

**(a) Bound classification:**

AI = 0.0833 FLOP/byte < I* = 31.25 FLOP/byte → **Memory-bound**

**(b) Attainable performance ceiling:**

```
Perf = min(P_peak, I x B_peak) = min(10,000, 0.0833 x 320) = 26.67 GFLOP/s
```

**(c) Architectural recommendation:**

Since vector addition is memory-bound, the bottleneck is DRAM bandwidth — not compute. The most impactful architectural change is **increasing DRAM bandwidth, for example by switching to HBM (High Bandwidth Memory)**, to raise B_peak. Adding more compute units would have zero effect since the kernel is already memory-limited.

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## Summary Table

| Kernel | FLOPs | Bytes | AI (FLOP/B) | Bound | Perf (GFLOP/s) | Recommendation |
|---|---|---|---|---|---|---|
| A — GEMM 1024x1024 | 2.147 G | 12.58 MB | 170.67 | Compute | 10,000 | Add MAC/ tensor cores |
| B — Vec Add N=4,194,304 | 4.19 M | 50.33 MB | 0.0833 | Memory | 26.67 | Switch to HBM |